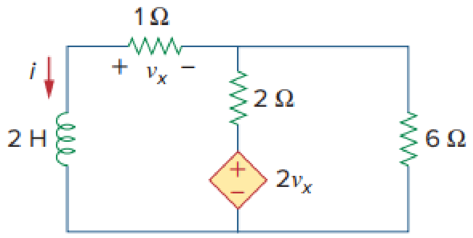
Chapter 7, Problem 3PP

Problem

Find i and vx in the circuit of Fig. 7.15. Let $i(0) = 7$ A.

Fig. 7.15:



Step-by-step solution

Step 1 of 5

Refer to Figure 7.15 in the textbook.

Redraw the circuit diagram with current notations as shown in Figure 1.

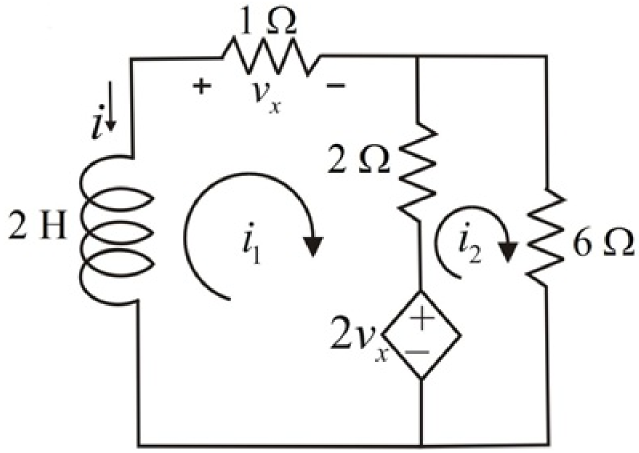


Figure 1

Step 2 of 5

Apply Kirchhoff's voltage law in loop 1.

$$2\frac{di_1}{dt}+(1)i_1+2(i_1-i_2)+2v_x=0$$

$$2\frac{di_1}{dt}+3i_1-2i_2+2v_x=0 \quad \text{......} \ (1)$$

Apply Kirchhoff's voltage law in loop 2.

$$-2v_x+2(i_2-i_1)+6i_2=0$$

$$-2v_x+8i_2-2i_1=0 \quad \text{......} \ (2)$$

From circuit diagram in Figure 2, the voltage v_x is as follows:

$$v_x=(1)i_1$$

$$=i_1 \quad \text{......} \ (3)$$

Step 3 of 5

Substitute i_1 for v_x in equation (1).

$$2\frac{di_1}{dt}+3i_1-2i_2+2i_1=0$$

$$2\frac{di_1}{dt}+5i_1-2i_2=0 \text{ ...... (4)}$$

Substitute i_1 for v_x in equation (2).

$$-2i_1+8i_2-2i_1=0$$
$$8i_2-4i_1=0$$
$$4i_1=8i_2$$

$$i_2=\frac{i_1}{2} \text{ ...... (5)}$$

Step 4 of 5

Substitute $\frac{i_1}{2}$ for i_2 in the equation (4).

$$2\frac{di_1}{dt}+5i_1-2\left(\frac{i_1}{2}\right)=0$$

$$2\frac{di_1}{dt}+5i_1-i_1=0$$

$$2\frac{di_1}{dt}+4i_1=0$$

$$\frac{di_1}{dt}=-2i_1$$

$$\frac{di_1}{i_1}=-2dt \text{ ...... (6)}$$

Integrate equation (6) on both sides.

$$\int_{i(0)}^{i(t)}\frac{1}{i_1}di=-2\int_0^t dt$$

$$\ln i_1\Big|_{i(0)}^{i(t)}=-2t$$

$$\ln\frac{i_1(t)}{i_1(0)}=-2t$$

$$\frac{i_1(t)}{i_1(0)}=e^{-2t}$$

$$i_1(t)=i_1(0)e^{-2t}$$

Since $i(t)=-i_1(t)$, therefore,

$$\left[-i(t)\right]=\left[-i(0)\right]e^{-2t}$$
$$i(t)=i(0)e^{-2t}$$

Substitute 7 A for $i(0)$.

$$i(t)=7e^{-2t}\text{ A}$$

Hence, the current through the inductor, $i(t)$ for $t>0$ is $\boxed{7e^{-2t}\text{ A}}$.

Step 5 of 5

The voltage across the $1\,\Omega$ resistor is as follows:

$$\begin{aligned} v_x(t) &= -[i(t)](1) \\ &= -i(t) \end{aligned}$$

Substitute $7e^{-2t}$ A for $i(t)$.

$$v_x(t) = -7e^{-2t}\ \mathrm{V}$$

Hence, voltage across the $1\,\Omega$ resistor, $v_x(t)$ for $t > 0$ is $\boxed{-7e^{-2t}\ \mathrm{V}}$.